Week 2 Assignment

R fundamentals and data wrangling

Complete this after Wednesday's tutorial. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 3's lecture (Monday). See the syllabus for the late policy.

Part 1: Functions and control flow

1. Write a function `classify_mass(x, threshold)` that returns `"heavy"` if `x > threshold` and `"light"` otherwise (default `threshold = 20`).
2. Create a `body_mass` vector with 6 values of your choosing and apply your function to it with `sapply()`.
3. Combine `body_mass` with a `species` vector (your choosing) into a tibble, and compute the mean mass per species.

Part 2: Data wrangling

Using the `ChickWeight` built-in dataset (`data(ChickWeight)`):

1. `glimpse()` and `summary()` the data.

What are the columns, and what type is each?

2. Filter to just `Diet == 1` and compute the mean `weight` at each `Time` point.
3. The data are already in long format (one row per chick per time point). Use `pivot_wider()` to reshape it so each chick is one row, with one column per time point.

What does a row represent now?

4. In 2–3 sentences: why would you want the long format for fitting a model, but the wide format might be more useful for, say, a table in a report?